PRAKTIKUM JARINGAN KOMPUTER

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E

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PERANCANGAN JARINGAN LABORATORIUM SEDERHANA MENGGUNAKAN PACKET TRACER

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MODUL 11

PERANCANGAN JARINGAN LABORATORIUM SEDERHANA MENGGUNAKAN PACKET TRACER

1.1 PENDAHULUAN

A. LATAR BELAKANG

Perancangan jaringan komputer merupakan hal yang sangat penting dalam sebuah instansi yang sudah menerapakan sistem komputer dalam pengelolaannya. Perancangan jaringan komputer harus sesuai dengan kebutuhan instansi terkait.Salah satu instansi yang banyak menerapkan jaringan komputer adalah di bidang pendidikan, terutama di universitas. Biasanya universitas menerapakan jaringan komputer untuk mengelola laboratorium sebagai penunjang proses belajar mengajar.Dalam modul terakhir ini, akan diterangkan contoh desain dan perancangan jaringan komputer untuk sebuah laboratorium komputer sederhana yang terdiri dari 3 sub jaringan (segmen),kemudian ke 3 segmen tersebut terhubung ke gateway.

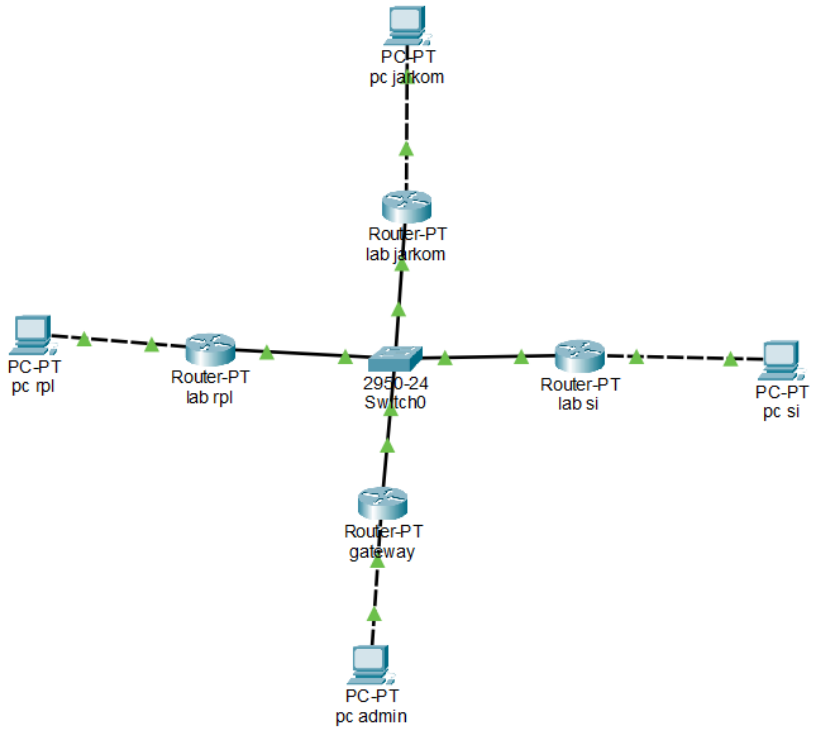
B. TUJUAN

Mahasiswa mampu mendesain simulasi jaringan sederhana untuk laboratorium computer setelah menerapkan konsep-konsep dari modul 1-9.

1.2 ALAT DAN BAHAN

1. Software Cisco.
2. Komputer.

1.3 KEGIATAN PRAKTIKUM



Gambar 11.1 Topologi praktikum

No	Nama Device	Interface	Alamat IP	Keterangan
1	Router Lab Jarkom	Ethernet 0	172.16.0.1/24	Ke PC Jarkom
		Ethernet 1	172.15.0.1/24	Ke 172.15.0.0
2	Router Lab Rpl	Ethernet 0	172.18.0.1/24	Ke pc rpl
		Ethernet 1	172.15.0.3/24	Ke 172.15.0.0
3	Router Lab SI	Ethernet 0	172.17.0.1/24	Ke pc si
		Ethernet 1	172.15.0.2/24	Ke 172.15.0.0
4	Router Lab Gateway	Ethernet 0	172.19.0.1/24	Ke pc admin
		Ethernet 1	172.15.0.4/24	Ke 172.15.0.0
5	PC Jarkom	Ethernet 0	172.16.0.2/24	Ke router jarkom
6	PC RPL	Ethernet 0	172.18.0.2/24	Ke router RPL
7	PC SI	Ethernet 0	172.17.0.2/24	Ke router si
8	PC Gateway	Ethernet 0	172.19.0.2/24	Ke 172.15.0.0

1.3.1 Lakukan konfigurasi Router:

lab jarkom

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname Jarkom
Jarkom(config)#int eth0
                      ^
% Invalid input detected at '^' marker.

Jarkom(config)#int eth0/0
%Invalid interface type and number
Jarkom(config)#int fa 0/0
Jarkom(config-if)#ip address 172.16.0.1 255.255.255.0
Jarkom(config-if)#no shut

Jarkom(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Jarkom(config-if)#ex
Jarkom(config)#int fa 1/0
Jarkom(config-if)#ip address 172.15.0.1 255.255.255.0
Jarkom(config-if)#no shut

Jarkom(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Jarkom(config-if)#ex
Jarkom(config)#
```

Gambar 11.2 Konfigurasi router jarkom

lab si

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname SistemInformasi
SistemInformasi(config)#int fa 0/0
SistemInformasi(config-if)#ip address 172.17.0.1 255.255.255.0
SistemInformasi(config-if)#no shut

SistemInformasi(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

SistemInformasi(config-if)#ex
SistemInformasi(config)#int fa 1/0
SistemInformasi(config-if)#ip address 172.15.0.1 255.255.255.0
SistemInformasi(config-if)#ip address 172.15.0.2 255.255.255.0
SistemInformasi(config-if)#no shut

SistemInformasi(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

SistemInformasi(config-if)#ex
SistemInformasi(config)#
```

Gambar 11.3 Konfigurasi router si

lab rpl

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname RPL
RPL(config)#int fa 0/0
RPL(config-if)#ip address 172.18.0.1 255.255.255.0
RPL(config-if)#no shut

RPL(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

RPL(config-if)#ex
RPL(config)#int fa 1/0
RPL(config-if)#ip address 172.15.0.3 255.255.255.0
RPL(config-if)#no shut

RPL(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
```

Gambar 11.4 konfigurasi router rpl

gateway

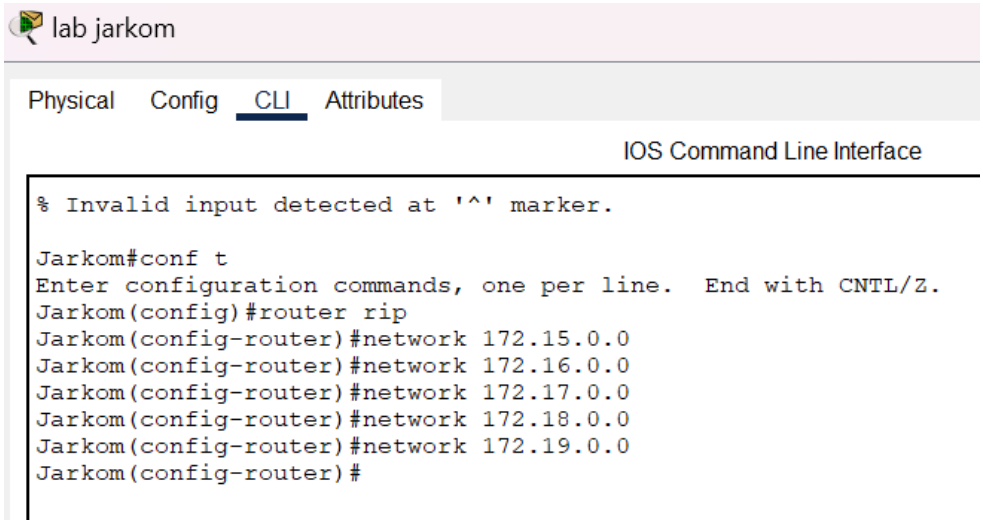
Physical Config CLI Attributes

IOS Command Line Interface

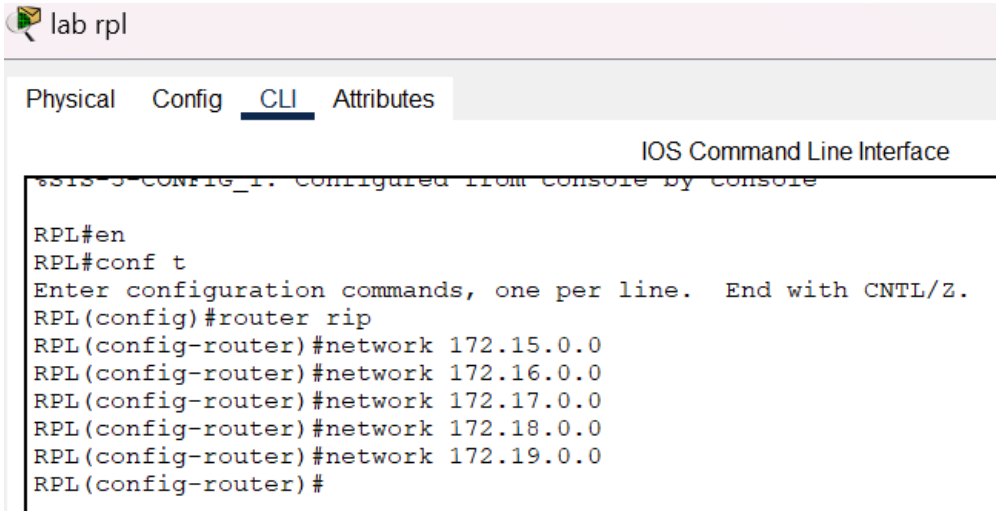
```
Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname UMS
UMS(config)#int fa 0/0
UMS(config-if)#ip address 172.19.0.1 255.255.255.0
UMS(config-if)#no shut

UMS(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

UMS(config-if)#int fa 1/0
UMS(config-if)#ip address 172.15.0.4 255.255.255.0
UMS(config-if)#no shut

UMS(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
```

Gambar 11.5 Konfigurasi router gateway

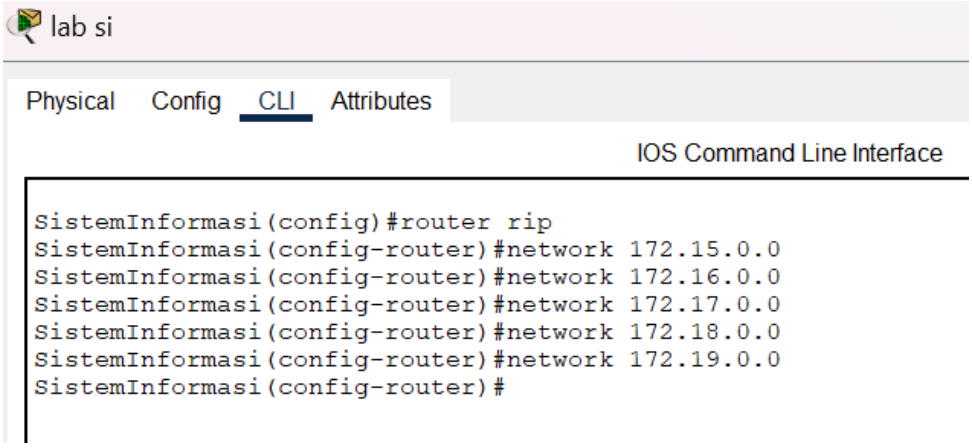
1.3.2 Lakukan konfigurasi rip routing:



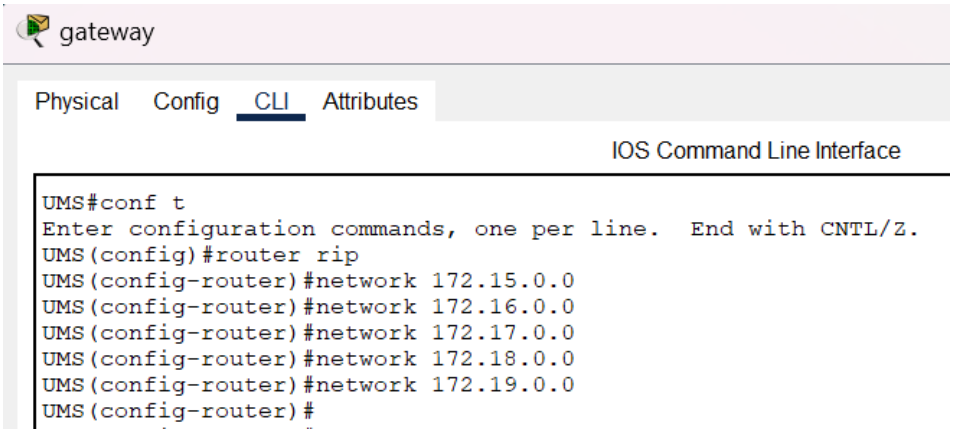
Gambar 11.6 Konfigurasi rip routing router jarkom



Gambar 11.7 Konfigurasi rip routing router rpl

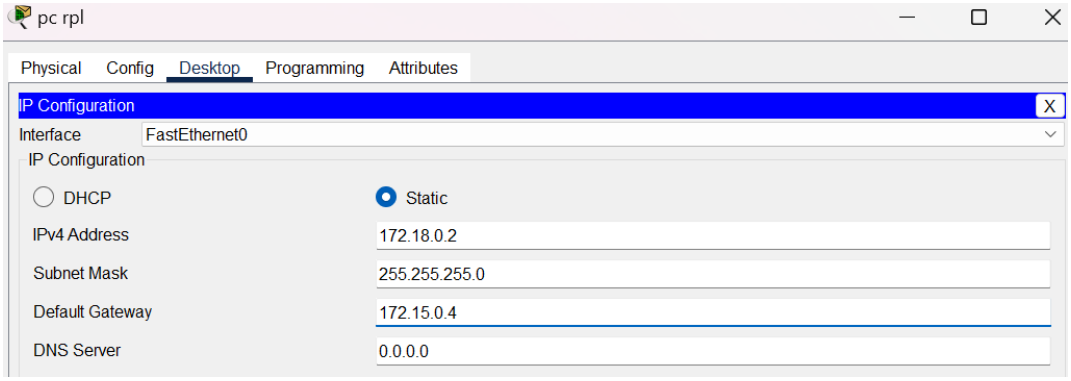


Gambar 11.8 Konfigurasi rip routing router si

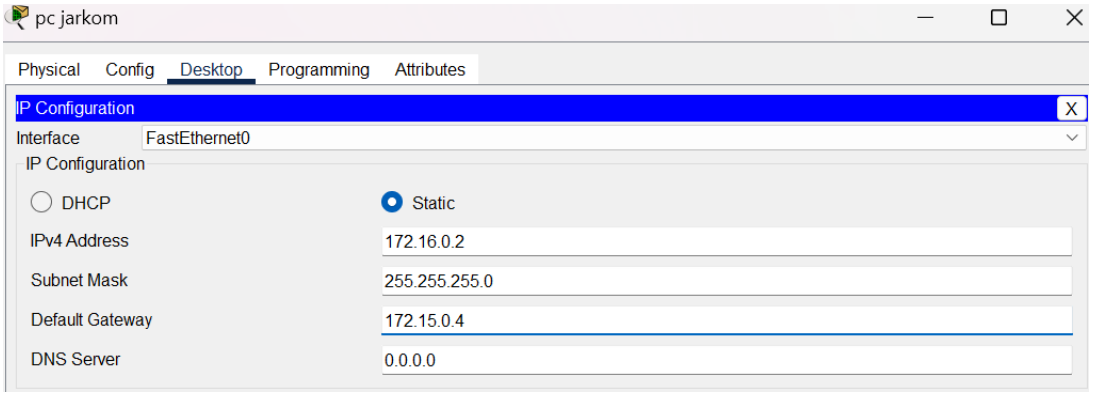


Gambar 11.9 Konfigurasi rip routing router gateway

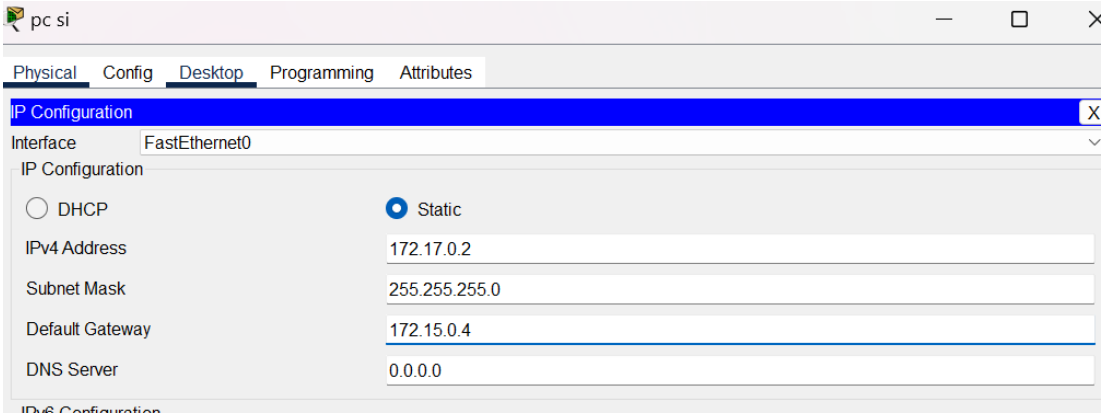
1.3.3 Konfigurasi IP masing-masing pc



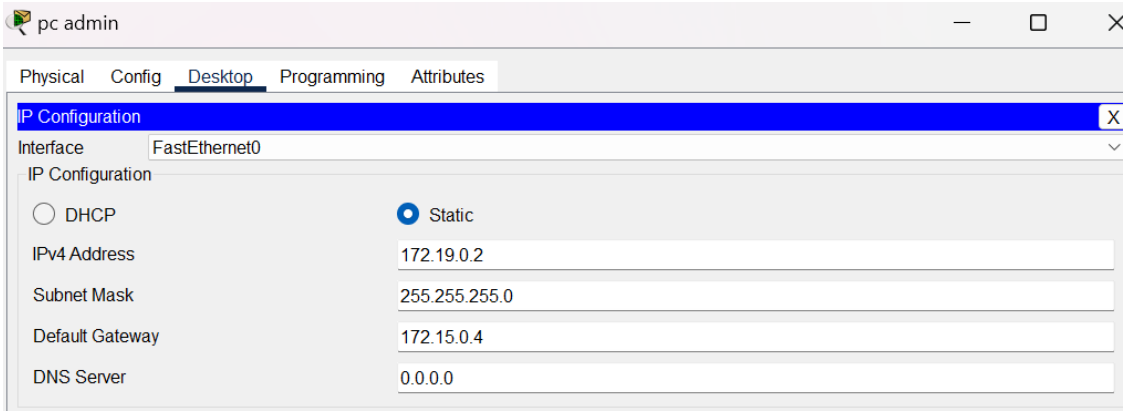
Gambar 11.10 konfigurasi IP pc rpl



Gambar 11.11 konfigurasi IP pc jarkom

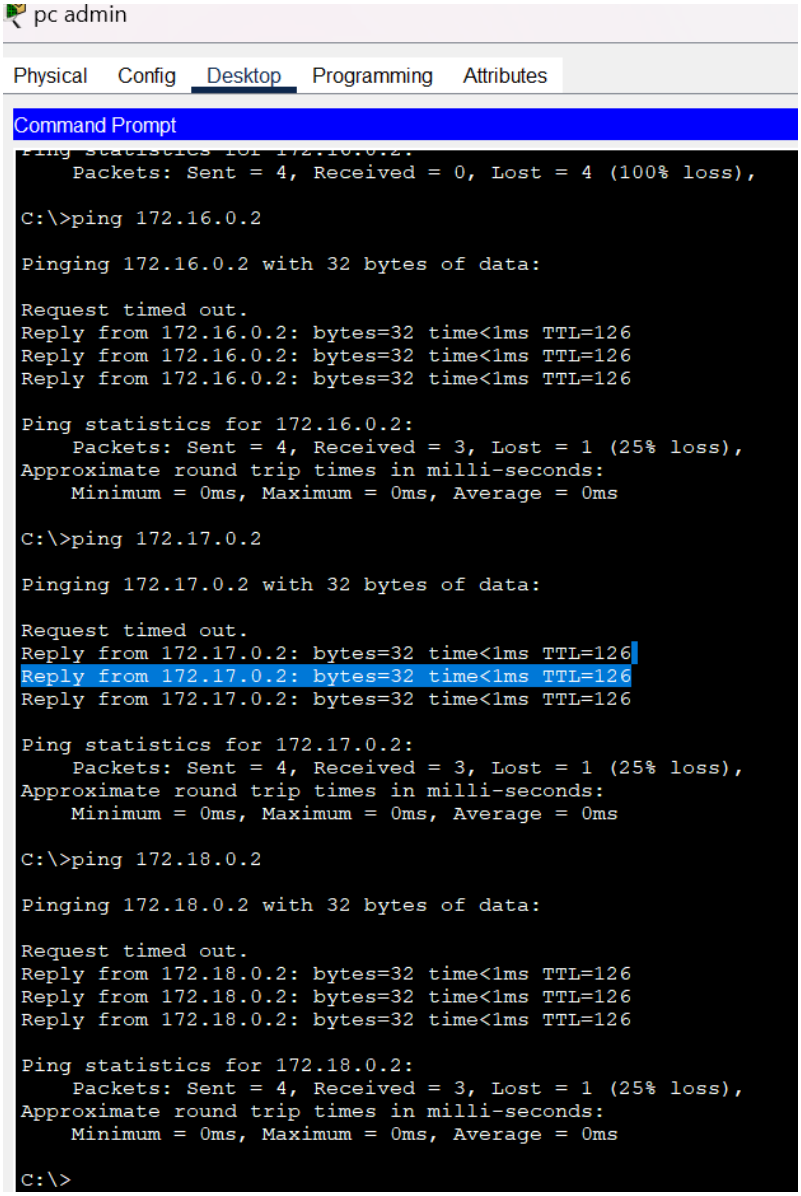


Gambar 11.12 konfigurasi IP pc si



Gambar 11.13 konfigurasi IP pc admin

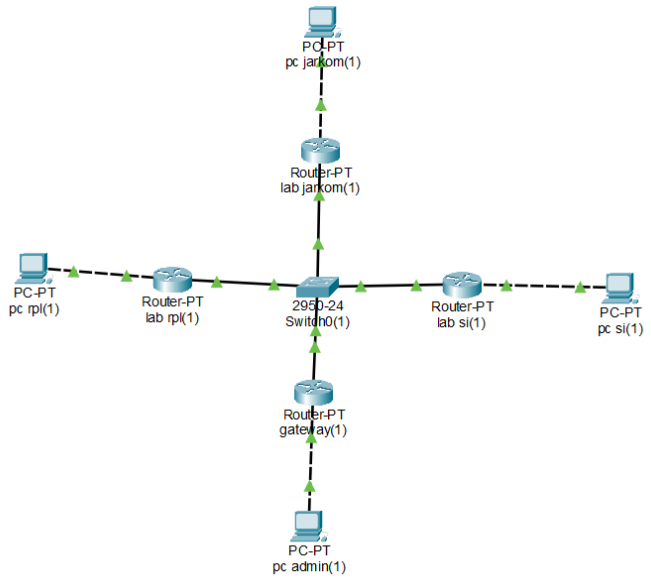
1.3.4 Melakukan pengujian ICMP request(ping)



Gambar 11.14 Pengujian dari pc admin ke semua pc

1.4 TUGAS

1.4.1.1 Buatlah topologi serupa, namun metode routing yang digunakan adalah routing statis.



Gambar 11.15 Topologi tugas

1.4.1.2 Konfigurasi

lab jarkom(1)

Physical Config CLI Attributes

IOS Command Line Interface

```
Jarkom>en
Jarkom#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Jarkom(config)#ip route 172.17.0.0 255.255.255.0 172.15.0.2
Jarkom(config)#ip route 172.18.0.0 255.255.255.0 172.15.0.3
Jarkom(config)#ip route 172.19.0.0 255.255.255.0 172.15.0.4
Jarkom(config)#ex
Jarkom#
%SYS-5-CONFIG_I: Configured from console by console

Jarkom#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.15.0.0/24 is subnetted, 1 subnets
C       172.15.0.0 is directly connected, FastEthernet1/0
     172.16.0.0/24 is subnetted, 1 subnets
C       172.16.0.0 is directly connected, FastEthernet0/0
     172.17.0.0/24 is subnetted, 1 subnets
S       172.17.0.0 [1/0] via 172.15.0.2
     172.18.0.0/24 is subnetted, 1 subnets
S       172.18.0.0 [1/0] via 172.15.0.3
     172.19.0.0/24 is subnetted, 1 subnets
S       172.19.0.0 [1/0] via 172.15.0.4

Jarkom#
```

Gambar 11.16 Konfigurasi routing dinamis pada router jarkom

lab rpl(1)

Physical Config CLI Attributes

IOS Command Line Interface

```
RPL>en
RPL#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RPL(config)#ip route 172.16.0.0 255.255.255.0 172.15.0.1
RPL(config)#ip route 172.17.0.0 255.255.255.0 172.15.0.2
RPL(config)#ip route 172.19.0.0 255.255.255.0 172.15.0.4
RPL(config)#ex
RPL#
%SYS-5-CONFIG_I: Configured from console by console

RPL#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.15.0.0/24 is subnetted, 1 subnets
C       172.15.0.0 is directly connected, FastEthernet1/0
     172.16.0.0/24 is subnetted, 1 subnets
S       172.16.0.0 [1/0] via 172.15.0.1
     172.17.0.0/24 is subnetted, 1 subnets
S       172.17.0.0 [1/0] via 172.15.0.2
     172.18.0.0/24 is subnetted, 1 subnets
C       172.18.0.0 is directly connected, FastEthernet0/0
     172.19.0.0/24 is subnetted, 1 subnets
S       172.19.0.0 [1/0] via 172.15.0.4

RPL#
```

Gambar 11.17 Konfigurasi routing dinamis pada router rpl

lab si(1)

Physical Config CLI Attributes

IOS Command Line Interface

```
SistemInformasi>en
SistemInformasi#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.15.0.0/24 is subnetted, 1 subnets
C       172.15.0.0 is directly connected, FastEthernet1/0
     172.16.0.0/24 is subnetted, 1 subnets
S       172.16.0.0 [1/0] via 172.15.0.1
     172.17.0.0/24 is subnetted, 1 subnets
C       172.17.0.0 is directly connected, FastEthernet0/0
     172.18.0.0/24 is subnetted, 1 subnets
S       172.18.0.0 [1/0] via 172.15.0.3
     172.19.0.0/24 is subnetted, 1 subnets
S       172.19.0.0 [1/0] via 172.15.0.4

SistemInformasi#
```

Gambar 11.18 Konfigurasi routing dinamis pada router si

gateway(1)

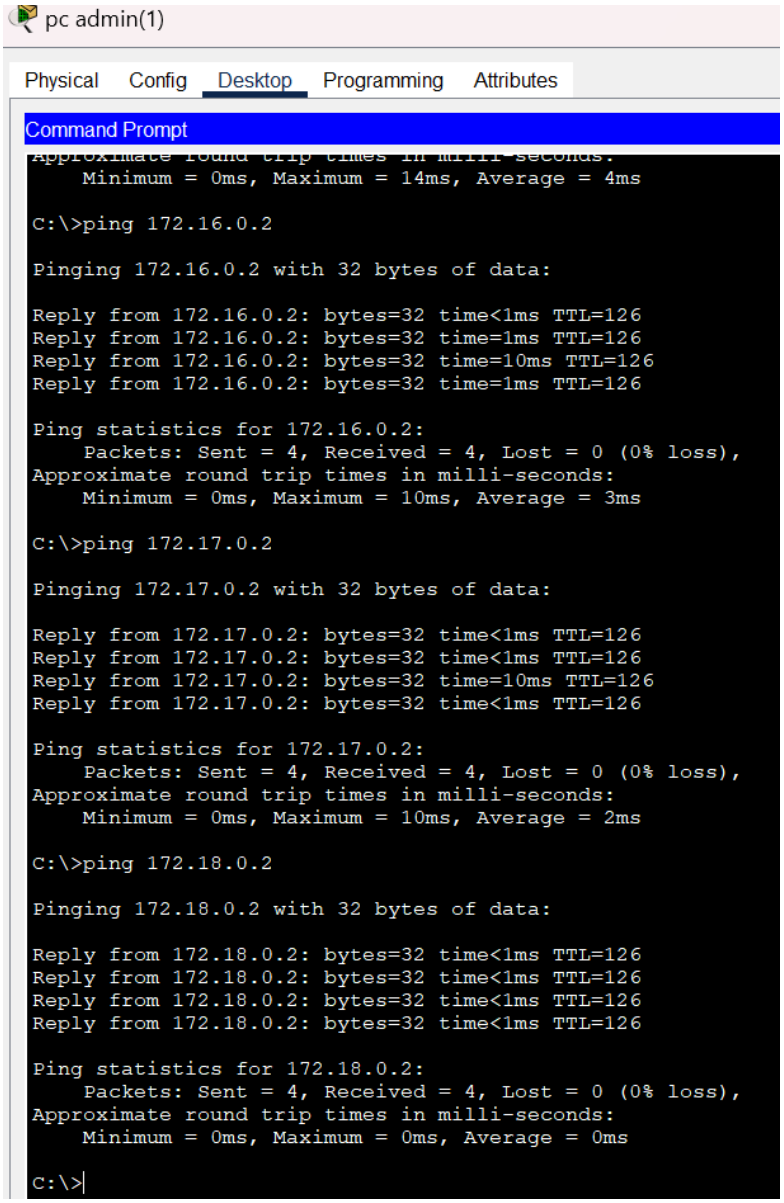
Physical Config CLI Attributes

IOS Command Line Interface

```
UMS>en
UMS#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
UMS(config)#ip route 172.16.0.0 255.255.255.0 172.15.0.1
UMS(config)#ip route 172.17.0.0 255.255.255.0 172.15.0.2
UMS(config)#ip route 172.18.0.0 255.255.255.0 172.15.0.3
UMS(config)#ex
UMS#
%SYS-5-CONFIG_I: Configured from console by console

UMS#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.15.0.0/24 is subnetted, 1 subnets
C       172.15.0.0 is directly connected, FastEthernet1/0
     172.16.0.0/24 is subnetted, 1 subnets
S       172.16.0.0 [1/0] via 172.15.0.1
     172.17.0.0/24 is subnetted, 1 subnets
S       172.17.0.0 [1/0] via 172.15.0.2
     172.18.0.0/24 is subnetted, 1 subnets
S       172.18.0.0 [1/0] via 172.15.0.3
     172.19.0.0/24 is subnetted, 1 subnets
C       172.19.0.0 is directly connected, FastEthernet0/0

UMS#
```

Gambar 11.19 Konfigurasi routing dinamis pada router gateway

1.4.1.3 Tes Ping



Gambar 11.20 PING pc admin ke semua pc

1.4.2.1 Buatlah topologi jaringan bus untuk membangun sebuah laboratorium computer yang terdiri dari 3 router dam berpusat pada 1 router gateway dengan metode routing statis dan dinamins

1.4.2.2 Topologi bus(Statis)

Gambar 11.21 Topologi BUS

1.4.2.3 Konfigurasi IP

gateway(2)

```
UMS#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
UMS(config)#int se2/0
UMS(config-if)#ip address 172.21.1.1 255.255.255.0
UMS(config-if)#clock rate 2000000
UMS(config-if)#no shut

%LINK-5-CHANGED: Interface Serial2/0, changed state to down
UMS(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
```

Gambar 11.22 Konfigurasi router gateway

lab rpl(2)

Physical Config CLI Attributes

IOS Command Line Interface

```
RPL#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RPL(config)#int se2/0
RPL(config-if)#ip address 172.21.1.2 255.255.255.0
RPL(config-if)#no shut

RPL(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

RPL(config-if)#ex
RPL(config)#int se3
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state
RPL(config)#int se3/0
RPL(config-if)#ip address 172.21.2.1 255.255.255.0
RPL(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
RPL(config-if)#
```

Gambar 11.23 Konfigurasi router rpl

lab si(2)

```
SistemInformasi#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SistemInformasi(config)#int se2/0
SistemInformasi(config-if)#ip address 172.21.2.2 255.255.255.0
SistemInformasi(config-if)#no shut

SistemInformasi(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

SistemInformasi(config-if)#ex
SistemInformasi(config)#int se3/0
SistemInformasi(config-if)#ip address 172.21.3.1 255.255.255.0
SistemInformasi(config-if)#no shut

%LINK-5-CHANGED: Interface Serial3/0, changed state to down
SistemInformasi(config-if)#
```

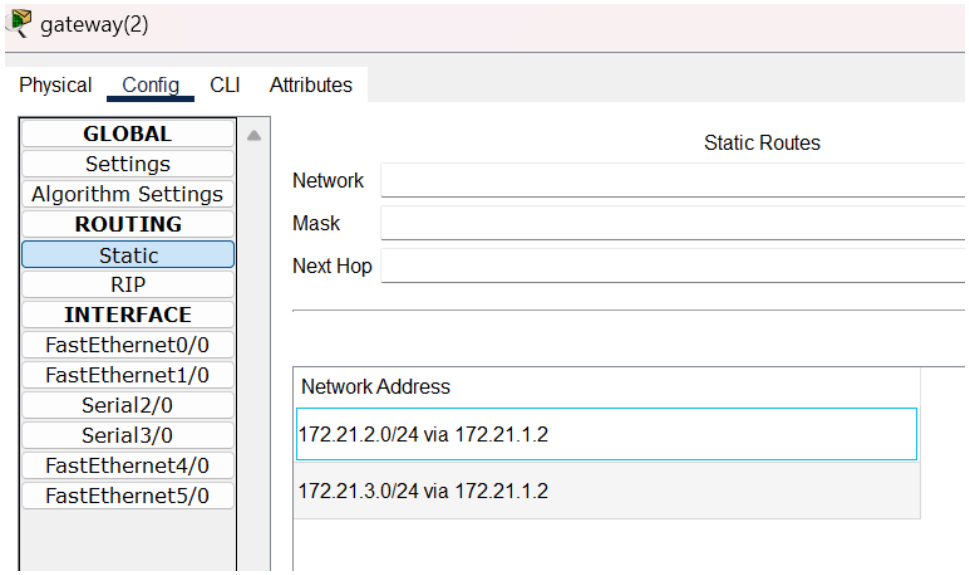
Gambar 11.24 Konfigurasi router si

lab jarkom(2)

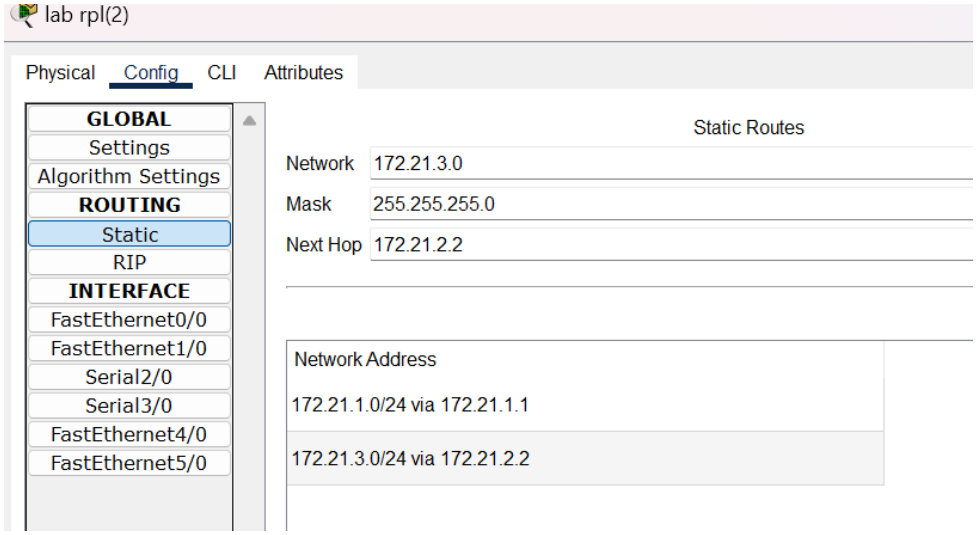
```
Jarkom>en
Jarkom#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Jarkom(config)#int se2/0
Jarkom(config-if)#ip address 172.21.3.2 255.255.255.0
Jarkom(config-if)#no shut

Jarkom(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Jarkom(config-if)#exit
Jarkom(config)#interface Serial2/0
Jarkom(config-if)#
Jarkom(config-if)#exit
```

Gambar 11.25 Konfigurasi router jarkom

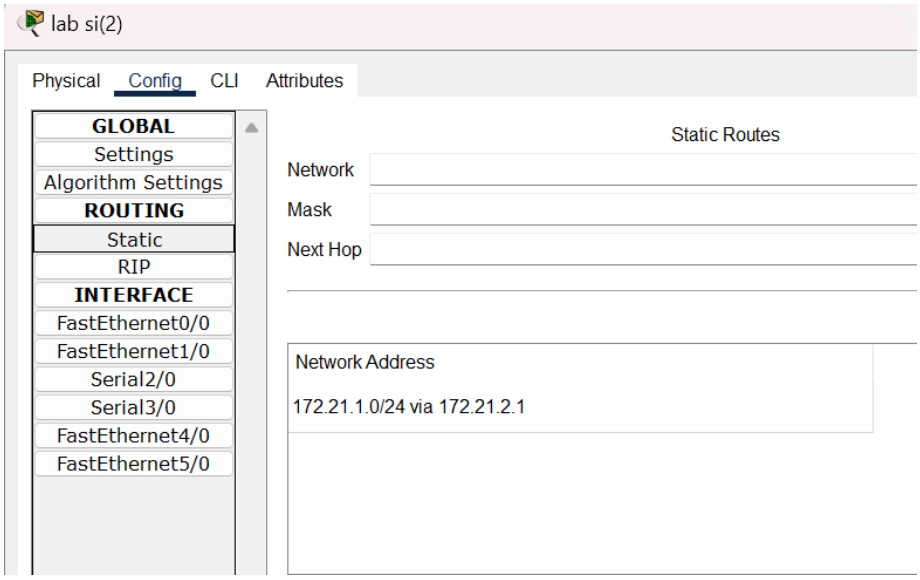
1.4.2.4 Konfigurasi Routing statis



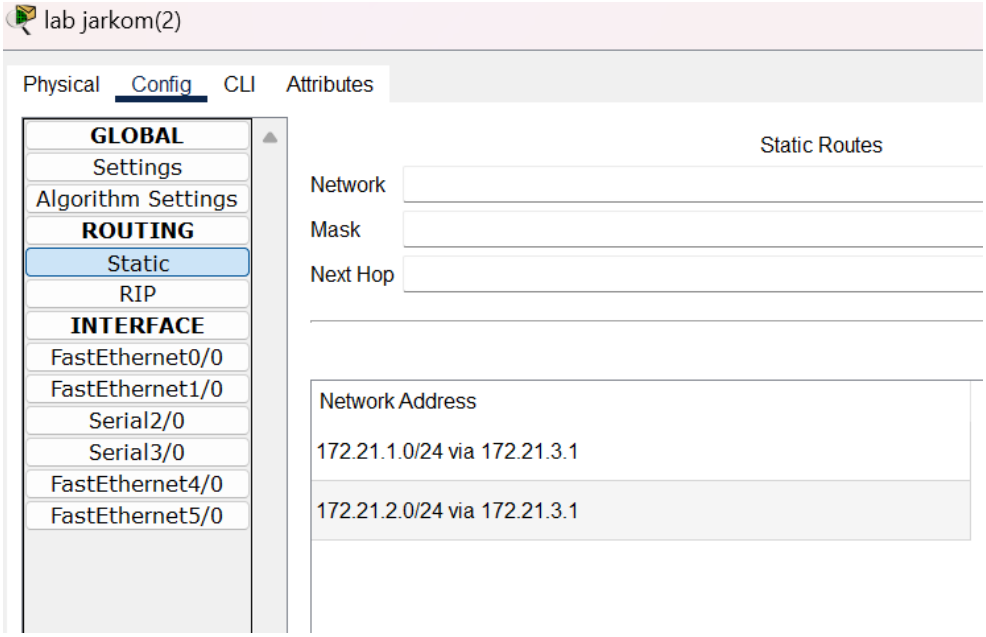
Gambar 11.26 Konfigurasi routing statis router gateway



Gambar 11.27 Konfigurasi routing statis router rpl



Gambar 11.28 Konfigurasi routing statis router si



Gambar 11.29 Konfigurasi routing statis router jarkom

1.4.2.5 Tes PING

gateway(2)

Physical Config CLI Attributes

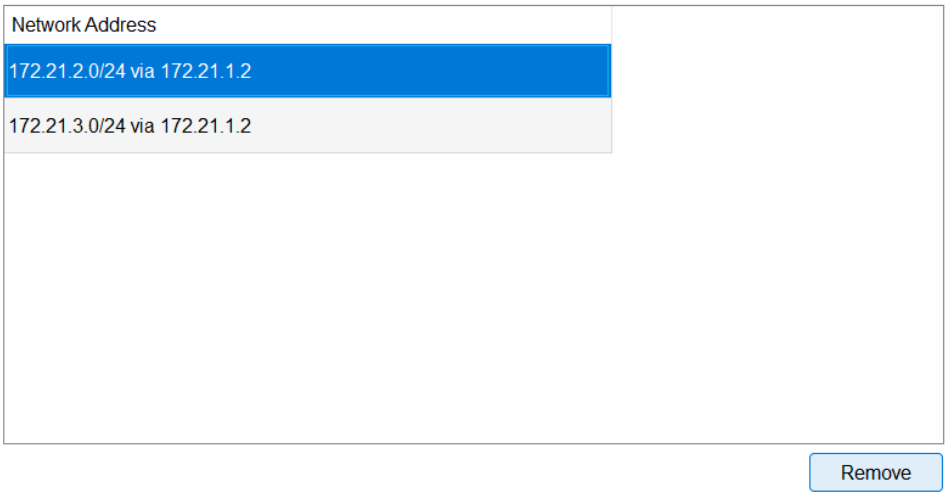
IOS Command Line Interface

```
UMS>ping 172.21.1.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.21.1.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 2/11/28 ms

UMS>ping 172.21.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.21.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 9/10/15 ms

UMS>ping 172.21.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.21.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 6/12/17 ms

UMS>ping 172.21.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.21.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 13/15/18 ms

UMS>ping 172.21.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.21.3.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 10/17/23 ms

UMS>
```

Gambar 11.30 Tes ping Router Gateway ke seluruh router

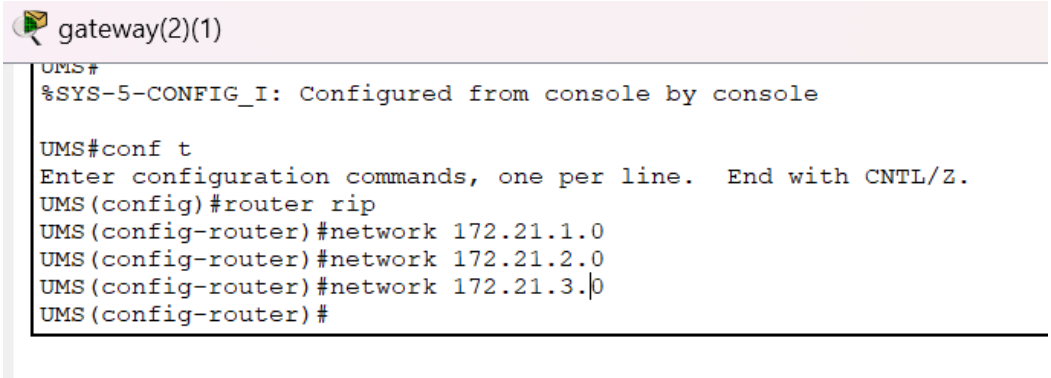
1.4.2.6 Topologi Bus (Dinamis)

Gambar 11.31 Topologi Bus Dinamis



Gambar 11.32 Menghapus konfigurasi routing static seluruh router

1.4.2.7 Konfigurasi Router RIP



Gambar 11.33 Konfigurasi router rip router gateway

lab rpl(2)(1)

```
RPL(config)#^Z
RPL#
%SYS-5-CONFIG_I: Configured from console by console

RPL#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
RPL(config)#router rip
RPL(config-router)#network 172.21.1.0
RPL(config-router)#network 172.21.2.0
RPL(config-router)#network 172.21.3.0
RPL(config-router)#
```

Gambar 11.34 Konfigurasi router rip router rpl

lab si(2)(1)

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#network 172.21.1.0
Router(config-router)#network 172.21.2.0
Router(config-router)#network 172.21.3.0
Router(config-router)#ex
Router(config)#
Router(config)#^Z
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 172.21.1.1
```

Gambar 11.35 Konfigurasi router rip router si

lab jarkom(2)(1)

```
Jarkom#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Jarkom(config)#router rip
Jarkom(config-router)#network 172.21.1.0
Jarkom(config-router)#network 172.21.2.0
Jarkom(config-router)#network 172.21.3.0
Jarkom(config-router)#
```

Gambar 11.36 Konfigurasi router rip router jarkom

1.4.2.8 Tes PING

gateway(2)(1)

```
UMS#ping 172.21.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.21.2.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/8/13 ms

UMS#ping 172.21.2.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.21.2.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 10/13/15 ms

UMS#ping 172.21.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.21.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/15/20 ms

UMS#ping 172.21.3.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 172.21.3.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 18/20/23 ms

UMS#
```

Gambar 11.37 Tes ping router Gateway ke seluruh router

DAFTAR PUSTAKA